Event-Local Sheafification and a Linear/Non-Linear Adjunction with Chu Coefficients

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Extended abstract — machine-checked in cubical Agda

Abstract

We construct an event-local sheafification functor and use it to lift a Benton linear/non-linear adjunction over a small site of finite causal interaction contexts. The objects of the site are finite local views of a concurrent interaction: unityped actors, the events in which they participate, and the causal order on those events. The topology is generated by decomposing each context into its single-event subcontexts. For any coefficient category with finite products, sheafification sends a presheaf to the product of its values on those subcontexts and is left adjoint to the inclusion of event-local sheaves. Given a Benton adjunction $F \dashv G$ between a cartesian coefficient category and a $*$-autonomous coefficient category, this construction yields an adjunction between cartesian-valued and linear-valued event-local sheaves. The cartesian side is the freely copyable-and-discardable regime, the $*$-autonomous side is the resource-sensitive regime with linear duals, and the induced comonad is the sheaf-level linear exponential "!". The hypotheses are also jointly realized in a concrete coefficient model: $\mathsf{V} = \mathrm{Chu}(\mathbf{Set}, \mathsf{Bool})$ has the required finite products and a minimal $*$-autonomous interface, and the Benton adjunction $\mathbf{Set} \rightleftarrows \mathrm{Chu}(\mathbf{Set}, \mathsf{Bool})$ is constructed in Agda. Separately, the postulate-free session syntax is interpreted into this Chu model by an underlying functor. Thus the sheaf-level theorem is generic, but not vacuous.

This note isolates one theorem from a larger programme that models concurrent interaction systems, and the typed data produced by them, in the language of sheaves and adjoint functors. The theorem starts from a deliberately austere view of agency. Actors are unityped: mathematically, they are elements of a bare set, and the formal system exposes no role, origin, or hidden state on which a construction can branch. The structure lies instead in events, causal dependence, incidence between events and actors, and the typed values assigned to finite contexts of interaction. The result below says that a coherent global assignment over such a context is freely reconstructed from its event-local assignments. The system's global point of view is not imposed from outside; it is the coherent reconstruction of its event-local standpoints.

The theorem has two parts. First, event-local observations freely generate a coherent global object. Second, that reconstruction respects the cartesian-linear structure of the values assigned to contexts: reusable data live on the cartesian side, resource-sensitive session-like data live on the linear side, and the Benton adjunction relates the two at the sheaf level.

The event-local site

The actor model of concurrency [1] describes computation as a collection of agents that interact solely by exchanging messages. We use only the unityped aspect of that tradition: actors are elements of a bare set, and the kernel exposes no role, origin, or tag on which a construction can branch. Lamport's causal order [2] supplies the temporal structure: events are partially ordered by possible influence. Following presheaf models of concurrency [3], these finite observations are organized as a small site.

Fix a set A of actors. A *finite causal context* over A consists of a finite set of events, a partial order on those events, and an incidence relation recording which actors participate in which events. The context is *supported*: every actor present in the context participates in at least one event. A morphism of contexts is a causality-reflecting embedding, preserving actors, events, and incidence, and both preserving and reflecting the causal order. Write Ctx_A for the resulting category; we call it the actor-causal site.

The topology used here is event-local. For an event e of a context F, write F_e for the single-event subcontext at e, whose actors are exactly those incident to e. The generating cover of F is

$$\{\, F_e \longrightarrow F \,\}_{e \in E(F)}.$$

No other covers are made primitive in this theorem. In particular, causal refinement, higher overlaps, and richer compatibility conditions are deferred to later topologies. The Agda development inherited the internal word "point" for these single-event subcontexts; in this abstract we use *event-local* for the site-side notion and reserve "points" and "states" for Chu spaces below.

Event-local sheafification

Let $\mathcal{C}$ be a coefficient category with finite products. A $\mathcal{C}$-valued presheaf on the site is a functor

$$P : \mathsf{Ctx}_A^{\mathrm{op}} \longrightarrow \mathcal{C},$$

and we write $\mathsf{PSh}_A(\mathcal{C}) = [\mathsf{Ctx}_A^{\mathrm{op}}, \mathcal{C}]$. An event-local sheaf is a presheaf satisfying descent for the single-event covers above. The formalization proves the expected product-locality characterization: a presheaf is event-local precisely when its value on a context is determined by its values on the single-event subcontexts of that context.

The inclusion $\iota_{\mathcal{C}} : \mathsf{Sh}_A^{\mathrm{ev}}(\mathcal{C}) \hookrightarrow \mathsf{PSh}_A(\mathcal{C})$ has an explicit left adjoint. For a presheaf P, define

$$a_{\mathcal{C}}(P)(F) = \prod_{e \in E(F)} P(F_e),$$

with restrictions induced by the corresponding maps between single-event subcontexts. This is sheafification in the restricted sense used here: the reflector onto presheaves satisfying descent for the explicitly presented event-local covers. It is not claimed to be the associated-sheaf functor for an independently formalized Grothendieck coverage. The machine-checked result is the adjunction

$$a_{\mathcal{C}} \dashv \iota_{\mathcal{C}}.$$

The linear/non-linear lift

Let C be a cartesian category and let V be a $*$-autonomous category [4], both with finite products. A linear/non-linear adjunction in the sense of Benton [5] is a monoidal adjunction $F \dashv G : \mathsf{C} \rightleftarrows \mathsf{V}$ from the cartesian world to the linear one. The induced comonad on V is the linear exponential "!". The Agda interface used by the theorem is intentionally minimal: it records the two functors, the adjunction, and the strong-monoidal comparison maps of F. The proof consumes the adjunction and product structure; the omitted global coherence and tensor-hom universal property are not unfolded.

Postcomposition lifts F and G to presheaves,

$$F_* : \mathsf{PSh}_A(\mathsf{C}) \to \mathsf{PSh}_A(\mathsf{V}), \qquad G_* : \mathsf{PSh}_A(\mathsf{V}) \to \mathsf{PSh}_A(\mathsf{C}),$$

with $F_* \dashv G_*$ computed pointwise. The right adjoint preserves event-local sheaves: descent for the single-event cover is transported across the hom-set bijection $\mathsf{C}(X, GY) \cong \mathsf{V}(FX, Y)$. The left adjoint is made to land in sheaves by composing with the reflector. With

$$L = a_{\mathsf{V}} \circ F_* \circ \iota_{\mathsf{C}}, \qquad R = G_*|_{\mathsf{Sh}^{\mathrm{ev}}_A(\mathsf{V})},$$

we obtain the theorem.

Theorem. *Let C be a cartesian category and V a $*$-autonomous category, both with finite products, and let $F \dashv G : \mathsf{C} \rightleftarrows \mathsf{V}$ be a Benton linear/non-linear adjunction. Then there is an adjunction*

$$L \dashv R : \mathsf{Sh}^{\mathrm{ev}}_A(\mathsf{C}) \rightleftarrows \mathsf{Sh}^{\mathrm{ev}}_A(\mathsf{V})$$

between cartesian-valued and linear-valued event-local sheaves on the actor-causal site. The induced comonad $L \circ R$ is the sheaf-level linear exponential.

A concrete Chu coefficient model

The theorem above is generic in its coefficient categories. The new Chu construction changes its status: the hypotheses are not merely assumed, but are simultaneously inhabited in a concrete machine-checked model. For a dualizing set k, a Chu object over **Set** is a triple $(A^+, A^-, \langle -, - \rangle : A^+ \times A^- \to k)$ [6, 7]. In the standard Chu terminology, A^+ is the carrier of points and A^- the carrier of states; this is distinct from the event-local subcontexts of the site.

At $k = \mathsf{Bool}$, the Agda development constructs the category $\mathrm{Chu}(\mathbf{Set}, \mathsf{Bool})$, its finite products, and the formalization's minimal $*$-autonomous interface. It also constructs the Benton adjunction

$$\mathbf{Set} \rightleftarrows \mathrm{Chu}(\mathbf{Set}, \mathsf{Bool}),$$

with right adjoint $G(A) = A^+$ and left adjoint $F(X) = (X, \mathsf{Bool}^X, \mathrm{ev})$. Supplying these data to the generic event-local theorem yields a concrete witness of the lifted adjunction $L \dashv R$. In this construction, the Benton linear/non-linear adjunction is the cartesian-linear bridge: it relates reusable cartesian values to resource-sensitive linear values. The Chu model supplies concrete coefficients in which that bridge exists, so the sheaf-level adjunction is an instantiated theorem rather than a purely conditional one.

Session syntax in the model

A separate result connects the session syntax to this coefficient model. The postulate-free syntax of session types and their structural morphisms, in the linear-propositions tradition for session types [8, 9], is interpreted into any target with the formalization's minimal $*$-autonomous and finite-product interface by an object map $J0$, a duality theorem $J0(S^{\perp_s}) \cong (J0(S))^{\perp}$, and a morphism map $J1$. Instantiating that interpretation at Chu($\mathbf{Set}, \mathsf{Bool}$) gives an underlying functor in the Cubical Agda category library from the session category to the Chu coefficient category. This is intentionally weaker than full session semantics: it is not an operational send/receive model, not a coherent monoidal or $*$-autonomous session functor, and not a construction of the relevant cyclic operadic action in Chu. The remaining action postulate in `Sessions.agda` is therefore not discharged by this result.

Scope and significance

Three classical lines meet in the theorem. It is a descent statement over a small site, in the tradition of Grothendieck [10, 11]. It is an instance of Lawvere's principle that adjointness organizes foundational structure [12]: the global object is obtained by a left adjoint, and its universal property is the freeness of the reconstruction. And it lifts Benton's linear/non-linear adjunction from coefficient categories to sheaves. The Chu instance adds non-vacuity: the general theorem is still stated parametrically, but there is now a concrete postulate-free coefficient model in which its hypotheses hold.

In the wider programme, this fixes a bounded piece of the operational side of a larger conjectural architecture. Event-local observations freely generate a coherent global object, that reconstruction is compatible with resource-sensitive values, and one concrete Chu coefficient model realizes the required value-side structure. Claims about richer causal descent, operational session semantics, formal introduction of external data into the system, and cross-site modal adjunctions belong to later stages.

Related work

The theorem has siblings by facet rather than a single direct predecessor. The local-to-global pattern is close to Abramsky and Brandenburger's sheaf-theoretic account of contextuality [13], where contextuality is detected as an obstruction to global sections, and to later cohomological developments of that line [14]. Event-based sheaf models of systems appear in work of Zardini, Spivak, Censi, and Frazzoli [15]. Local views of distributed computation are organized topologically in work of Goubault, Ledent, and Rajsbaum [16].

On the linear-concurrency side, thin concurrent games and generalized species provide a recent adjacent model [17], while work on session types derived from linear logic clarifies the session-theoretic side [9, 18]. Licata, Shulman, and Riley's fibrational framework for substructural and modal logics [19], and the modal homotopy type theory of Riley, Finster, and Licata [20], are further siblings on the type-theoretic side: they organize movement between logical regimes rather than collapsing those regimes into one. These works share pieces of the structure below, but do not construct the actor-causal site, event-local sheafification, Benton lift, and Chu coefficient instance proved here.

Formalization

The formalization splits into three compact layers. The role of Agda here is not only to record the statement: it checks that the reflector, lifted adjunction, finite-product witnesses, and Chu instance inhabit the same interfaces simultaneously.

- The event-local reflector and lifted adjunction are formalized without postulates in:

  ```
  SupportedChangeOfBase.agda
  SupportedLinearNonlinearSheafAdapter.agda
  PointSheafAdjunction.agda
  PointLinearNonlinearAdjunction.agda
  ```

- The concrete Chu stack is in:

  ```
  Chu.agda
  ChuFiniteProducts.agda
  ChuMonoidal.agda
  ChuLinearNonlinear.agda
  ChuPointVProd.agda
  ChuPointExample.agda
  ```

  The final Agda witness is named `Chu-nonvacuity`; its fully expanded type is omitted because writing it out makes the type-checker expand definitions prohibitively.

- The session interpretation is in:

  ```
  SessionSyntax.agda
  SessionMorphismSyntax.agda
  SessionObjectModel.agda
  SessionChuModel.agda
  ```

Limitations

We are explicit about what is not done here.

- The value-level adjunction $F \dashv G$ is assumed in the general theorem; it is discharged only in the concrete Chu instance $(\mathbf{Set}, \mathrm{Chu}(\mathbf{Set}, \mathsf{Bool}))$.

- The Chu model uses the formalization's minimal $*$-autonomous interface. The global monoidal coherence and tensor-hom universal property of Chu categories are not fully formalized here.

- The topology only imposes descent along single-event covers. Causal order, overlaps of events, and richer compatibility conditions do not yet carry descent content.

- Event-local sheaves are defined by descent for the explicitly presented single-event covers. We do not yet compare this with a fully formalized Grothendieck coverage or library sheaf condition for the same site.

- The Sessions-to-Chu interpretation is an underlying functorial realization of the syntax, not operational send/receive semantics and not a coherent monoidal or $*$-autonomous session semantics.

- This is not a cross-site modal adjunction, not a construction of the required cyclic operadic action in Chu, and not an account of the formal introduction of external data into the system.

A note on authorship

This abstract was drafted with the assistance of a language model and reviewed by the authors; the surrounding framework and its philosophical motivation are presented separately in the project repository at `https://codeberg.org/doc2cog/hoapa`. The formal claims reported here are backed by proofs that the cubical Agda type-checker accepts in the modules named above, with the scope limitations stated explicitly.

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